EPICS® SYLLABUS

COURSE & PROGRAM WEBSITE: EPICS Home Page

Address: Armstrong Hall of Engineering (ARMS), Room 1200, 701 W. Stadium Ave., West Lafayette, IN 47907-2045

For general questions, email: Indianapolis: epics-ind@purdue.edu, West Lafayette: epics-wl@purdue.edu.

COURSE INFORMATION

- This syllabus covers all courses with the EPCS prefix
- The CRNs for the courses can be found at Register - EPICS - Purdue University
- Meeting times can be found at Schedules - EPICS - Purdue University. Consult the schedule for times for online synchronous courses.
- Instructional modality: in person with some asynchronous (online) assignments
- Course credit hours: Variable: 1-3 credits
- Course Brightspace pages vary by section; visit Brightspace Homepage - Purdue West Lafayette.

EPICS PERSONNEL

ADMINISTRATIVE STAFF

Staff can be reached via email. For general questions, contact the Indianapolis staff at epics-ind@purdue.edu and West Lafayette at epics-wl@purdue.edu.

ADVISORS

Instructors in EPICS are called *Advisors*. Their role is to coach and mentor the teams on their projects, to assess student progress, and assign grades. Your advisor is indicated as the instructor in myPurdue and also in Brightspace. Your advisors might be part of the EPICS administrative or full-time instructional staff, Purdue faculty, or industry volunteers.

TAs

A Teaching Assistant (TA) is assigned to each EPICS team; however, all TAs have office hours to work with students from any team. Collectively, they form a TA Consulting Pool from which any student or team is encouraged to seek expertise relevant to their project. The office hour schedule can be found at: EPICS schedules

UGTAs

Undergraduate TAs (UGTAs) are available at selected times to assist students working in the lab/makerspace. Working hours are posted throughout the facilities, or you can contact the corresponding space manager for details.

ADDITIONAL RESOURCES

For computer and information technology support email: IT@purdue.edu or visit Purdue IT homepage

COURSE DESCRIPTION

EPICS is a community-engaged design course in which teams of students from across campus work together on long-term projects that benefit the community. Project work centers around the engineering, technology, and computing needs of a community partner, but interdisciplinary team interaction is an integral element for project success.

Students may participate in EPICS multiple semesters; participation for multiple consecutive semesters on a project team is encouraged. Teams are composed of first year students through seniors.

Most EPICS projects last at least one year; however, partnership with the community organization continues for several years. Projects are intended to solve real problems, are defined in partnership with their community partners, and span the complete design process cycle (i.e., problem identification – specification development – conceptual design – detailed design – delivery – service/maintenance – retirement).

You receive academic credit for participating in EPICS. How academic credits are applied to your major depends on your degree program and is determined by your department and/or advisor. Visit How EPICS counts at Purdue.

IN EPICS, YOU WILL LEARN AND EXPERIENCE:

Interdisciplinary Design: Learn how to be better designers; gain design knowledge and skills; learn how to apply disciplinary knowledge to real and possibly ill-defined problems; learn how to identify and acquire new knowledge; learn to collaborate with people from other disciplines and develop an appreciation for cross-disciplinary contributions in design.

Professional Preparation: Develop a broad set of skills needed to be successful in the ever-changing global workplace and professional job market.

Community-Engaged Learning: Provide significant service to the community while learning; gain an understanding of the role that engineering (and other disciplines) can play in society and the broader issues related to the needs we are addressing.

SPECIFIC COURSE LEARNING OBJECTIVES ARE AS FOLLOWS:

- **Discipline Knowledge:** ability to apply concepts from your major to the design of community-based projects
- **Design Process:** an understanding of design as a start-to-finish process
- **Lifelong Learning:** an ability to identify and acquire new knowledge as a part of the problem-solving/design process
- **Customer Awareness:** being mindful of the person requesting the project (i.e., client, partner)
- **Teamwork:** an ability to function on multidisciplinary teams and an appreciation for the contributions from individuals from different fields of study
- **Communication:** an ability to communicate effectively, both orally and written, with people from widely varied backgrounds
- **Ethics:** an awareness of professional ethics and responsibility
- **Social Context:** an appreciation of the role that your discipline can play in social contexts

GRADING

In EPICS, students work on teams, and their final individual grade will reflect the quality and quantity of the student's documented:

- Learning and skill development per the course learning objectives
- Individual accomplishments
- Team accomplishments

For non-Senior Design students, the "Individual Evaluation Rubric" provides an overall description of the characteristics of the different grade levels of individual work, and it can be found at:

Individual Evaluation Rubric - EPICS - Purdue University

Grading guidelines for Senior Design Students can be found by visiting:

Senior Design Verification Process in EPICS at Purdue

EPICS students may receive grades of A+, A, A-, B+, B, B-, C+, C, C-, D+, D, D- or F.

All team members are responsible for the progress of the project. Teams will work together to identify team project goals for the semester. Individual roles and responsibilities within the team and projects will be identified. Formal assessment of the learning and accomplishments will be done at mid-semester, and again at the end of the semester for final grading. Evaluation will be based on the components listed in the individual evaluation rubric, including project progress, communication with project partner, individual accomplishments and documentation, lab and lecture attendance, project management and design documentation, design review presentations and documentation, social media posting, and informal lab presentations and demonstrations. Senior design students will also be graded on their specific requirements. Input into the assessment decisions will be collected from the project partner, advisors, TAs, and the team members themselves. Each student will be asked to critically assess his/her own participation in the project and that of all team members (peer evaluations). Students will be allowed to propose modifications to the responsibilities if appropriate.

The mid-semester assessment is a formative assessment intended to provide more detailed and cumulative feedback on the learning and accomplishments to date in the semester. General feedback may be provided at the mid-semester, and if desired by the student or required by advisor, at other times. Project documentation and team artifacts will be assessed during the semester per the Milestones - EPICS - Purdue University.

ATTENDANCE POLICY AND TIME REQUIREMENTS

Students must attend all the two-hour lab sections each week throughout the semester, either virtually or in person. In accordance with Purdue's attendance policy, and in consideration of the myriad issues that may arise in the lives of students, we recognize that it may be necessary for a student to be absent. When the absence can be anticipated, the student must notify their advisor and team as far in advance as possible. In the case of emergency absences (e.g., acute illness or family emergency), students must contact their advisor and team as soon as feasible.

For cases that fall under Purdue's excused absence regulations (i.e., bereavement, military service, jury duty, parenting leave, or medically excused absence policy for students), visit the ODOS' Class Absences website to complete the appropriate request forms. ODOS reviews the requests and, when granted, will notify your instructors.

Students are responsible for any work they miss due to their absences and are expected to minimize the impact on the team project work. Only advisors can excuse a student from a course requirement or responsibility. Unexcused absences will negatively impact the student's course grade. The University expects students and their instructors to approach problems with class attendance in a reasonable way. Additional meetings and work times are to be scheduled by the project team members. Students are expected to attend those meetings.

The typical expectations regarding time spent on EPICS are as follows, but like other courses on campus, you may need to spend more time than is typical to achieve a similar outcome:

- **2 credit hours** = average 5.0 hours/week **outside** the lab on your EPICS project and/or learning activities
- **1 credit hour** = average 3.5 hours/week **outside** the lab on your EPICS project and/or learning activities

Professional Development Hours (PDHs) are typically 50-minute-long skill sessions designed to introduce skills useful to project progress and the development of course-related outcomes. PDHs requirements vary based on the number of credit hours you are registered for and if you are a new or returning student to EPICS.

- Students enrolled for **1 credit hour** must accumulate at least **five (5) PDHs**.
- Students taking two or three credit hours must have a semester total of 10 PDHs.
- There are five (5) prescribed PDHs for new students**.**
- PDHs, including lecture series and skill sessions, will be held throughout the semester, to allow returning students and students registered in two or three credits to fulfill the PDH requirements.

- Students may select other PDHs with the approval of their advisor.
- Participation will count towards the requirement as specified in the Learning Activity or Skill Session description.

SEMESTER COURSE WORK

An overview of the semester activities can be found in the Milestone Schedule on the EPICS website. Individual and team assignments are found in Brightspace. In addition, the website contains templates and guidelines for the course documents listed below.

In the event of a major campus emergency or crisis, course requirements, deadlines, and grading percentages are subject to changes that may impose a revised semester calendar or other circumstances beyond the instructor's control. EPICS will work to keep information flowing via email (see the list of contacts at the top of this syllabus) and on the EPICS website: EPICS Homepage - Purdue. Monitor your @purdue.edu email regularly.

ALL STUDENTS:

- **Individual Evaluation Rubric:** self-assessment tool to document individual contributions and learning; it also facilitates the advisor's evaluation. The document is completed by the student at mid-semester and end of semester.
- **Individual Documentation:** all students are required to maintain individual documentation to demonstrate their weekly individual contributions and thinking via an online notebook. All activities related to the project, including individual efforts and ideas, reflections, relevant material and discussions from lectures, contacts, team sessions, conversations and meetings with the project partner, are to be date-recorded or referenced in the individual documentation. Individual documentation will be reviewed as indicated on the Milestones Schedule.
- **Peer Evaluation:** Students will complete a peer evaluation of their project team members at mid-semester and at the end of the semester using CATME.
- **Final Reflection:** Individuals are to complete a final reflection at the end of the semester.

SENIOR DESIGN STUDENTS (ADDITIONAL REQUIREMENTS):

Students MUST obtain explicit approval from the team advisor before participating as a senior design student.

- **Senior Design Project Proposal:** Must be completed during the first semester of Senior Design
- **Senior Design Project Description**: Must be completed during the second semester of Senior Design.
- **Documentation**: The senior design project and individual student outcomes must be documented carefully throughout the senior design experience.
- **Senior Design Outcomes Matrix**: An index of how the course outcomes have been met over the year and where evidence for this mastery can be found (notebook, project documentation, etc.). It is also used by the advisor(s) and EPICS admin to approve the satisfaction of course outcomes.
- **Senior Design Rubric:** Senior design grading will be completed in accordance with the Senior Design Outcome Rubric, found at Senior Design - EPICS - Purdue University.

TEAMS:

- **Team Responsibilities:** Teams will document team roles in Brightspace and Microsoft Teams. Those can be updated by the TA with the support of the Team Leader/Project Manager.
- **Design Document:** Each project has a *design document* which provides a comprehensive and detailed description of the project design, as well as the project management information. It also contains the *project charter*, which establishes the goals of the project and the overall timeframe. The intended audience is future

EPICS teams who need to continue the design in subsequent semesters or teams that need to maintain and/or repair the project after it is fielded. The project management information includes the *semester plan* and *transition report*. The design document should be seen as a "living" document and posted on the team's Microsoft Teams site.

- **Design Reviews:** Formal reviews by people external to the team completed in mid-semester and at the end of the semester to provide additional feedback on projects. Teams must prepare documentation prior to the review to send to the reviewers. After the review, teams are responsible for documenting what issues were raised and how they were addressed (some may be resolved in the following semester).
- **Delivery Review:** The delivery review will be held with the Design Reviews for projects to be delivered this semester. A *delivery checklist* must be completed and returned no later than week 14.
- **Social Media:** Teams must show regular progress in the EPICS LinkedIn page; use @EPICS at Purdue to tag and augment the reach of your team's development and news.

ACADEMIC INTEGRITY AND PROFESSIONAL RESPONSIBILITY

EPICS expects every member of the Purdue community to practice honorable, ethical, and professional behavior both inside and outside the classroom. In EPICS, students are encouraged to work together and share information. When indicated, students and teams can modify previous versions of documents to be submitted for the current assignment. However, it is unacceptable for students to claim individual work that is not their own or to use sources without appropriate citation. It is also unacceptable for students to misrepresent information to their instructional staff, their team, and/or their community partner**. The use of large language models (LLM), generative artificial intelligence, or any other means of producing documentation that is not authentically written by the students in their own words for notebooks, reflections, design or transition documents, or any other course document is strictly prohibited.** In addition, misuse of EPICS resources is considered dishonest. At the instructor's discretion, instances of academic dishonesty will result in a reduced score, a zero score, or a failing grade for the course. All occurrences of academic dishonesty will be reported to the Office of Students Rights and Responsibilities (OSSR) with a copy to their school. If there is any question as to whether a given action might be construed as academic dishonesty, please see your team's advisor or TA before you engage in any such action.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breeches of this value by either emailing integrity@purdue.edu or by calling (765)49-48778. While information may be submitted anonymously, the more information that you provide the greatest opportunity for the university to investigate the concern. For current Purdue policies, visit the Academic Integrity and Academic Regulations & Student Conduct - Purdue University pages.

Purdue Honor Pledge:

"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

ACCUTE OR CHRONIC ILLNESS

If you are sick with an acute or chronic medical condition, please contact the Purdue University Student Health (PUSH) patient portal (Home - Student Health Web Portal). If you find yourself too sick to progress in the course, notify EPICS via email or Brightspace. The Office of the Dean of Students (odos@purdue.edu) is also available to support you should this situation occur.

STUDENTS WITH DISABILITIES

Students with all ranges of abilities are welcome and encouraged to participate fully in all aspects of the course. If you feel you may need accommodation based on the impact of a disability, you should contact your team advisor privately to discuss your specific needs. Also, the Disability Resource Center (DRC) can assist in coordinating reasonable accommodations for students with documented disabilities at the following locations:

- Indianapolis – Engineering and Technology Building (ET) room 331
- West Lafayette – Ernest C. Young Hall (YONG) room 830

Students may present a "Letter of Accommodation" at any point in the semester.

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let your advisor and/or EPICS administration know so that we can discuss options. You are also encouraged to contact the DRC at: drc@purdue.edu or by phone: 765-494-1247.

NONDISCRIMINATION STATEMENT

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. The University believes that intellectual and cultural diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Purdue's nondiscrimination policy can be found at Nondiscrimination Policy Statement - Purdue University.

HEALTH & WELLNESS

MENTAL HEALTH

Your mental wellbeing is important. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, contact Counseling and Psychological Services (CAPS) at (765)49-46995 and Home CAPS during and after hours, on weekends and holidays. CAPS counselors are located at the following sites during business hours:

- Indianapolis – Engineering and Technology Building (ET), suite 324 (enter through ET 331)
- West Lafayette – Purdue University Student Health (PUSH)

Additional resources can be found at ODOS Campus Resources, Therapy Assistance Online, and CARES.

BASIC NEEDS SECURITY

If you are facing challenges securing basic needs such as food, housing, transportation, health services, or access to technology or childcare resources, and believe this may affect your performance in the course, please contact the Office of the Dean of Students (ODOS) to help coordinate with Community Resources. These services vary by location. In West Lafayette, see the Basic Needs Program website, or email basicneeds@purdue.edu. To connect with a Student Support Generalist on the Indianapolis campus, contact them by phone at 765-495-7797 or email studentlifeindy@purdue.edu.

EPICS RESOURCES

EPICS Project Work Locations:

Location	Room Name	Building and Room No.
Indianapolis	Maker Space* (shared with other engineering programs)	IO 115
West Lafayette	EPICS Design Build Lab*	ARMS B103
West Lafayette	EPICS Computer Lab*	ARMS 1095

Location	Room Name	Building and Room No.
West Lafayette	EPICS Electronic & Computing Hardware Lab* EPICS Project Prototyping Lab*	ARMS 1098

*All students are expected to review the *Lab Safety Awareness and Use Guidelines* and complete the online form annually in Brightspace to receive access to EPICS lab facilities during evening and weekend hours via PUID card. **Forms must be submitted via Brightspace no later than the end of the first week of the semester.** More information about the EPICS lab spaces can be found at: EPICS Labs - Purdue University.

The EPICS lab facilities in ARMS (West Lafayette) are for EPICS students only. EPICS students may forfeit access privileges by allowing access to other students without prior consent from the Lab Manager. The Maker Space in IO 115 (Indianapolis) is a shared space for students in the College of Engineering. Please be respectful and allow equal access to its resources.

All students and teams are responsible for keeping their project areas clean.

EPICS students are expected to use the equipment in the labs responsibly. Abuse of the lab facilities may result in loss of access, a failing grade for the course, and/or formal disciplinary action.

COMPUTER ACCOUNTS

TEAM COMPUTING RESOURCES

Microsoft Teams is an integrated environment that allows collaboration and content management. Each EPICS team has its site where they can store and edit documents, assign tasks, have discussion groups, chats, etc.

Although Teams is the primary method of communication, each team also has an email list service. The format is: *epics-@ecn.purdue.edu* (e.g., epics-ims@ecn.purdue.edu for the IMS team). Please note the team email is originally set up for team members only, not the advisor or TA. The advisor and/or TA can be added if desired.

At the discretion of the advisor, the team can also maintain a web home page. The URL would be http://epics.ecn.purdue.edu/xxxxx, where xxxxx is the team acronym (e.g., epics.ecn.purdue.edu/ISD for the ISD team). The team home pages are also linked to the main EPICS home page.

OTHER RESOURCES:

EPICS has resources available to support your project work (e.g., laptops, reference books, video cameras, measurement devices). Please contact the EPICS Lab Manager to check out those resources.

ADDITIONAL INFORMATION

DISCLAIMER

This syllabus is subject to change. You will be notified of any changes as far in advance as possible via announcements on Brightspace, Teams, or email. Monitor those platforms regularly for updates and notifications.

EMERGENCY PREPAREDNESS

Please subscribe to Purdue ALERT text messages and familiarize with Purdue's emergency procedures guide flipbook at: Emergency Procedures Guide - Preparedness and Planning - Purdue University. Please review the emergency procedures included below for Indianapolis and the West Lafayette Neil Armstrong Hall of Engineering (ARMS).

EMERGENCY PROCEDURES

Fire & Emergencies, Call or Text: 911
Purdue University Police: (765) 494-8221

SEVERE WEATHER

- Calmly proceed to the basement level of the building.
- Lecture Halls 1010, B061, & B071 are preferred for the occupants to gather.

MEDICAL EMERGENCY

- Avoid leaving the injured person except to summon help.
- Do not move the injured person.
- Render FIRST AID or CPR if you are trained and feel comfortable doing so.
- Call or text 911.

FIRE

- Pull the nearest fire alarm and use stairwells to leave the building.
- Do not re-enter building until directed by emergency personnel.
- Alert other people as you evacuate and provide assistance to others if necessary.
- Gather to the west of Armstrong Hall in Emergency Assembly Area.

ACTIVE THREAT

- Seek immediate shelter, preferably in a room that can be locked, barricaded, or secured.
- Remain in place until emergency personnel provide additional guidance or tell you it is safe.

INDIANAPOLIS

EMERGENCY PROCEDURES

PURDUE UNIVERSITY

TORNADO

SEEK SHELTER

- Move to an interior room on the lowest level.
- Stay away from windows and exterior doors.
- Listen to a weather radio or local news for updates.
- Stay away from hazardous materials.

FIRE

EVACUATE

- Pull the fire alarm.
- **CALL 911.**
- Leave the building and **CLOSE DOORS**.
- If unable to exit the building, go to the nearest stairwell or place of refuge. Do not use elevators.
- Assemble in a designated area.

ACTIVE SHOOTER

RUN - HIDE - FIGHT

- **CALL 911.**
- Leave the building.
- If evacuation is not possible, hide in a concealed place.
- Lock and barricade door, turn off the lights.
- Wait for law enforcement.
- As a **LAST RESORT**, overpower the shooter.

MEDICAL EMERGENCY

STAY CLEAR

- **CALL 911.**
- Provide the location, nature of injury or illness, current condition of the victim and other requested information.
- Do not move the victim unless in immediate danger.
- If trained, administer first aid or CPR.

911 EMERGENCY: POLICE/FIRE/MEDICAL

FOR MORE INFORMATION:

protect.iu.edu

NON-EMERGENCY

- IUPD-Indianapolis 317-274-2058
- Campus Facilities Services 317-278-1900
- Environmental Health & Safety (EHS) 317-274-2005
- IU Emergency Management & Continuity 317-274-8152
- Loss Control (INLOCC) 317-274-8951

UTILITY FAILURE

IDENTIFY LOCATION

- Call Campus Facilities Services or IUPD to report issues.
- Be prepared to provide failure type and location.
- University officials may evacuate a building due to utility failures.

SUSPICIOUS ACTIVITY

SEE SOMETHING, SAY SOMETHING

- Examples include a person is behaving strangely, an object is out of the ordinary and there is a gut feeling that something is wrong.
- If something is spotted, **CALL 911.**

HAZARDOUS MATERIALS

STAY BACK

- If life threatening, pull the fire alarm, evacuate, and **CALL 911.**
- If not life threatening, call EHS (IUPD for nights and weekends) and provide information on type of incident and location.
- If contact is made with a hazardous material, decontaminate and seek medical attention as necessary.

BOMB THREAT

REMAIN CALM

- Get as much information as possible from the caller – location of device, what it looks like, what will cause it to explode.
- Note background sounds, gender of caller and any notable characteristics of the caller's voice.
- **CALL 911.**

During an emergency, always CALL 911 FIRST.

TERMS TO KNOW:

Human-Centered Design Process	Process designers follow in developing a viable, feasible, and desirable solution in which the people impacted by the design are considered throughout the development.
Design Documentation	All information vital to the project which would allow any new project member to determine why the project was started, what decisions/reasoning were made along the way, what specifications and restrictions were imposed on the project; that is, any information that would allow them to continue where it was left off or to completely recreate the project.
Design Lead (Project Leader)	Role that oversees the project design and responsible for facilitating the project through all aspects of the design process.
EPICS	Formerly *Engineering Projects In Community Service*; as an established interdisciplinary program, the acronym is better known and preferred.
Financial Officer	Team role to manage team's budget and expenses.
Individual Evaluation Rubric (IER)	Evaluation tool used to document individual contributions and learning and facilitate both self- and advisor assessment.
Lab/lecture	The "lab" portion of the EPICS course is the 2-hour block designated in student's schedule to meet with the whole team and to work on their projects.
Outcome	One of the eight broad learning goals of the EPICS courses that students who successfully participate in the course over multiple semesters will have demonstrated.
Peer Evaluation	It allows students to assess themselves and their peers. It is completed at mid-semester and at the end of the semester. See "Semester Course Work" section for more info.
PDH (Professional Development Hour)	45 to 60minute long lecture or skill session designed to introduce knowledge useful to project progress and the development of course related outcomes.
Project	An individual engineering/technology/computing need that an EPICS team hopes to meet.
Project Manager (Team Leader)	Team role responsible for the overall operation and effectiveness of the team and provides planning, direction, and guidance.
Project Partner	A community organization that communicates an engineering need that a team is created to meet in continued coordination with the EPICS program and students.

Project Partner Liaison	Team member selected to be the main point of contact between the team and the project partner.
Project Evaluation Rubric	Evaluation tool used to document project progress and accomplishments and facilitate both self- (project) and advisor assessment.
Role	Word used to describe one of the positions held by students on each team.
Senior Design	Intensive engineering design experience required for all senior engineering students to graduate. The EPICS coursework currently can fill this requirement for ECE, MDE and CS students. See “Semester Course Work” section for more info.
SharePoint	SharePoint is an integrated environment that allows collaboration and content management. Each EPICS team has its site where they can store and edit documents, assign tasks, have discussion groups, and create wikis. It can be accessed via Teams.
Skill session	A learning activity intended to help develop skills and knowledge needed for project work and related course outcomes. Counts towards required number of PDHs.
Web (epics) and Team File (epics-shares) Shares	Physical server space hosted by the former engineering computer network (ECN, now Purdue IT) and used in the past for web files and collaboration. Nowadays, only historical videos and photos are maintained and may be accessible only through ECN web share (epics). Approval may be required to access the old Web and Team File Share content when necessary.
Team (section)	Group of students in the same section, partnering with the same organization(s), and sharing lab time (e.g., BME, EWB, IS); the terms Project Team or sub-team are sometimes used to describe the group of people working together on a specific project.
Communication Officer	Team member selected to post updated team member information and progress on social media; the individual also leads team recruiting.